MIT.nano at the Massachusetts Institute of Technology certifies that

Paulina Lilian Ruíz Servín

*has successfully completed the **2025** NanoLab, a one-week hands-on introduction to microfabrication technology using the diverse toolsets of MIT.nano. Part of the Monterrey Tec-MIT Nanotechnology Program, the NanoLab covers a wide range of devices, describing their fundamentals, fabrication process flow, and data analysis.*

October 27 – October 31, 2025
Massachusetts Institute of Technology

LUIS VELÁSQUEZ-GARCÍA
Principal Research Scientist, Instructor
Microsystems Technology Laboratories
Massachusetts Institute of Technology

VLADIMIR BULOVIĆ
Monterrey Tec-MIT Nanotechnology Program Director
Fariborz Maseeh (1990) Chair in Emerging Technology
Massachusetts Institute of Technology

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The Code-optional Data Contest for Everyone! The Contest is Now Closed

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AND THE WINNERS ARE...



1st PRIZE: Customized, Emotion-Driven Music Selection Using Facial Expression Recognition

Nirlesh Saravanan, from Sri Sairam Engineering College, delivers a compelling first-prize project that seamlessly connects human emotion with artificial intelligence through music.

Built entirely in **Altair AI Studio**, his emotion-based music recommendation system combines facial-expression analysis and deep learning to create a personalized, mood-aware listening experience. Using **DeepFace** to detect emotions such as happiness, sadness, or neutrality, the system interprets real-time facial cues with high accuracy. A second deep learning model analyzes detailed audio features from thousands of songs to classify music by emotional tone and recommend a matching Spotify track.

By integrating computer vision, audio analytics, and intelligent workflow design, this project demonstrates how AI can translate human signals into adaptive, emotionally responsive digital experiences in real time.

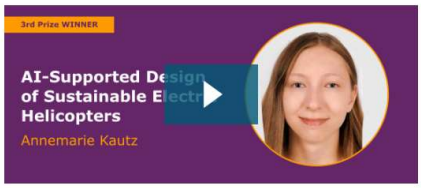


2nd PRIZE: Improving Air Quality through Better, More Reliable Monitoring and Causation Discovery

Swietenia Naomi Medina Gasca from Anáhuac Cancún University takes a thoughtful, data-driven approach to air-quality monitoring by examining not pollution itself, but the strategy behind how monitoring networks are designed.

She chose to work in the 3rd contest category, in which she used the Altair-provided dataset but came up with her own problem statement. Using **Altair AI Studio**, she transformed the complex, unstructured data into a clean analytical workflow, built a predictive model, and visualized key patterns to reveal how station placement, pollutant focus, altitude, landform, and urban context shape monitoring coverage.

Her work demonstrates how AI can validate existing monitoring strategies and inform smarter, more effective network expansion—highlighting the power of data science to support environmental decision-making beyond simple measurement.



3rd PRIZE: Designing Electric Aircraft Through AI Input with Simulation and ChatGPT-Generated Data

Annemarie Kautz from TU Munich, working with an international student team from **Tongji University**, demonstrates how AI can dramatically accelerate electric aircraft design.

Using simplified eVTOL models, simulation data, and AI-expanded datasets, the team explored how design decisions impact flight time, sustainability, and material use—revealing clear performance-sustainability trade-offs. With an A/B/C rating system and classification models built in **Altair AI Studio**, they created a faster, more transparent design workflow, including lifecycle tracking through a battery passport.

The project was inspired by **Professor Juergen Grotepass**, whose long-standing commitment to international, high-impact engineering education continues to motivate students to fuse speed, quality, and sustainable innovation in aerospace design.

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Honorable Mentions

Predicting Coffee Flavor Using Measurable Quality Factors

Sarah Dambach, Honor Henry, Seth Green, Gavin Cooper, Fairfield University - USA

Coffee quality is often judged by tradition and taste, but this team brought data to the table. Using Altair AI Studio, they analyzed characteristics like aroma, acidity, and moisture to build models that predict overall quality and even forecast flavor.

Which measurable factors matter most? Can flavor be predicted before the first roast? Their insights reveal the strongest predictors of flavor and give producers, traders, and certifiers a data-backed way to improve quality and forecast customer satisfaction.

A crisp, data-driven take on what makes great coffee.

Watch the video here.

Improving Bridge Safety Through Predictive Maintenance

Xijia Wang, Jilin University - China

Bridge maintenance is critical, yet traditional methods struggle to keep pace. This project uses Altair AI Studio to build a smarter, data-driven system that predicts maintenance priority, repair type, and cost using classification and regression models trained on historical data.

By analyzing how these targets relate, the solution uncovers patterns that help engineers act earlier and more accurately. With standardized data and Explainable AI, the insights remain transparent and reliable.

The results: improved safety, reduced failure risk, longer bridge life, lower costs, and more sustainable infrastructure planning. A proactive step toward resilient cities.

Watch the video here.

AI-Driven Data Analysis for Vehicle Energy Efficiency

José Diego González Fernández, Paulina Lilian Ruiz Servín, ITESM - México

In the Shell Eco Marathon, every bit of efficiency counts, and this team decided to let data lead the way.

By capturing performance signals from their vehicle and analyzing them in Altair AI Studio, they cleaned the data, tested predictive models, and built their own to uncover which factors genuinely shape energy use during each run.
The result? Clear insights to guide the design of next season's car and fresh ideas that could influence how energy-efficient vehicles are built in the real world.

From lowering consumption to reducing CO_2 emissions, their project shows how smart data analysis with Altair AI Studio can drive meaningful impact. A sharp, data-driven look at what makes a vehicle truly efficient.

Watch the video here.

Faculty Impact Acknowledgement

Dr. Vishnu Vinekar, Professor of Analytics, Dolan School of Business at Fairfield University - USA

We are pleased to recognize Dr. Vishnu Vinekar for his outstanding influence and dedication to student engagement in this year's contest.

Through his mentorship and encouragement, more than 20 student teams submitted projects, representing the highest level of participation driven by a single faculty member. One of his student teams also received an honorable mention for its contribution.

Dr. Vinekar's commitment to applied, project-based learning exemplifies the vital role faculty play in inspiring students to explore AI, data-driven problem solving, and real-world innovation beyond the classroom.

Certificate of Completion

Presented to

Paulina Lilian Ruíz Servín

FOR COMPLETION OF THE ALTIUM EDUCATION
PCB BASIC DESIGN COURSE

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22 October 2026

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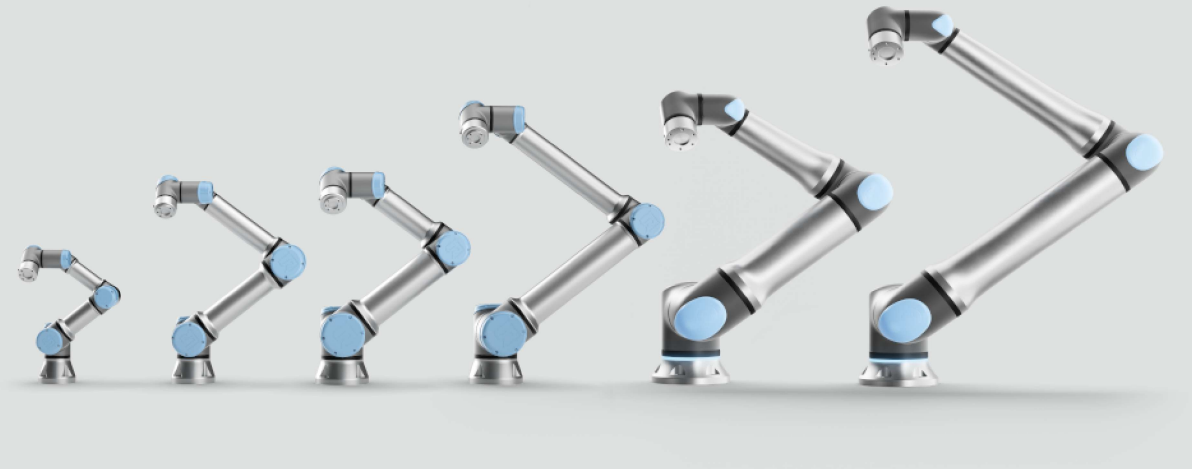
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e-Series Core Track

Paulina Lilian Ruiz Servin

2. March 2026

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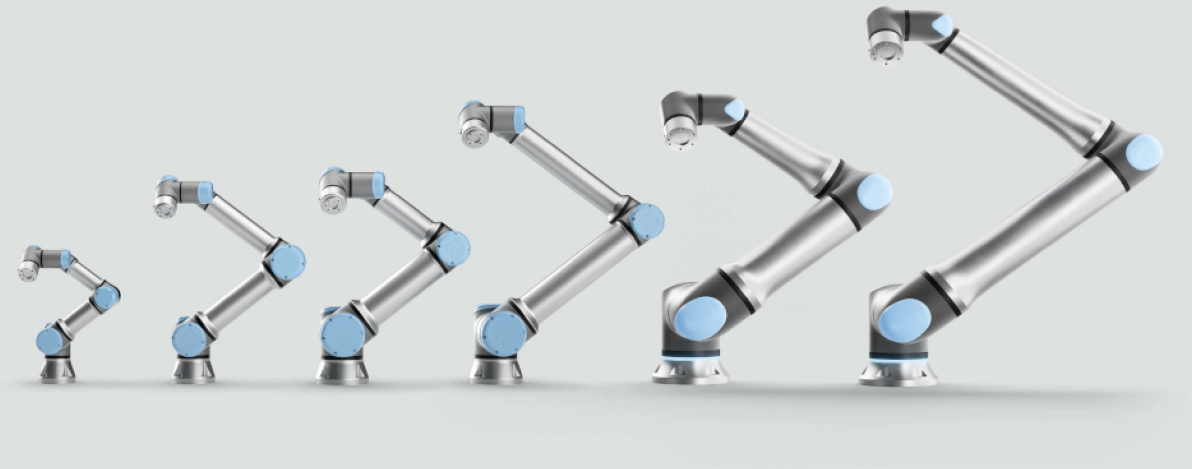
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e-Series Pro Track

Paulina Lilian Ruiz Servin

2. March 2026

Martin Woersaa Abildgaard
Universal Robots Academy



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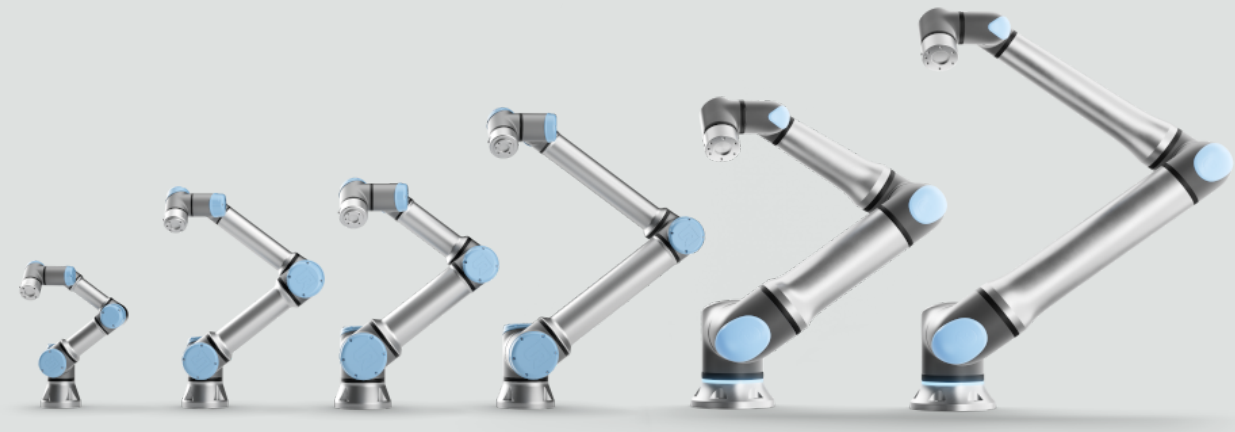
Certificate of completion

e-Series Application Track

Paulina Lilian Ruiz Servin

19. October 2026

Martin Woersaa Abildgaard
Universal Robots Academy



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Certificate of completion

Educational Robotics Training - Core

Certificate number: E202603074593780

School name: Tec de Mty Campus CDMX

Paulina Lilian Ruiz Servin

13. March 2026

Universal Robots Academy



Curriculum

Paulina Lilian Ruiz Servin 13. March 2026

Module 1: e-Learning
- Online e-Series Core Track Modules 1-8

Module 2: Safety
- Basic knowledge about Robotic environments as it pertains to safety
  - Safety Sensors
  - Situational Awareness
  - Interactions between the robot and its surroundings

Module 3: Intro to Robotics
- Basic knowledge on the History of Robotics
- Traditional vs. Collaborative Robots
- An understanding of Robotics and Industry 4.0

Module 4: The Robotics World
- Basic knowledge on the different categories of Robots
- An understanding on the different types of Industrial Robots
- Understanding different applications / uses for Robots in different industries

Module 5: Getting Started
- Moving the robot by using the “Move” tab
- Knowledge about the I / O Connections and how-to setup
- How to configure the TCP, Payload, and Center of Gravity
- How to navigate the Polyscope GUI

Module 6: Operations and Motion
- Basic knowledge about singularities
- The different movement types and how to use them
  - MoveJ
  - MoveL
  - MoveP
  - MoveC
- How to configure a blend radius
- Configuration of speed and acceleration of movements and individual waypoints
- Apply skills acquired previously to an actual robot
  - Programming a Pick-and-Place application

Module 7: Pick and Place
- Types of Inputs and Outputs (I/O)
- How to wire up I/O
- Building a simple Pick and Place Application
- Understanding Pick, Place, Transit, Approach, and Exit Locations

Module 8: Programming
- Configuration and use of if-else commands
- Basic knowledge about variables
- How to program a subprogram
- Configuration and usage of switch-case commands

Module 9: Palletizing
- Configuration and usage of the palletizing template

Module 10: Features / Planes
- How to create a coordinate system (plane)
- How to program relative to a coordinate system

Module 11: Safety Settings
- Configuration and use of the available safety settings
  - Robot limits
  - Joint limits
  - Safety planes
  - Tool position
  - Direction
  - Configurable Inputs / Outputs
  - Safe Home

UNIVERSAL ROBOTS
A Teradyne Robotics Company

Paulina Lilian Ruiz Servin
has successfully completed

Personnel Safety

Issue date 21. October 2026



Louise Loehndorff
Mobile Industrial Robots Academy

Paulina Lilian Ruiz Servin
has successfully completed

Autonomous Navigation

Issue date 22. October 2026



Louise Loehndorff
Mobile Industrial Robots Academy

Paulina Lilian Ruiz Servin
has successfully completed

Mission Programming

Issue date 22. October 2026



Louise Loehndorff
Mobile Industrial Robots Academy

SIEMENS

Xcelerator Academy

Paulina Lilian Ruiz Servin

is certified as

Process Simulate Standalone Associate

Ashley Hodge
Siemens Xcelerator Academy

Issue Date: Jun 12, 2026

This certificate confirms that the person named above successfully completed the exam for the stated certification.

DASSAULT SYSTEMES

SOLIDWORKS

CERTIFICATE

Dassault Systèmes confers upon

PAULINA LILIAN RUÍZ SERVÍN

the certificate for

SOLIDWORKS Design Associate

(CSWA)



August 19 2026

Manish KUMAR
SOLIDWORKS CEO
R&D Vice President

C-DAB8S6Z3UL

Shell Eco-marathon

Americas 2025

Off-Track Award Certificate

Presented to:

EcoVolt CCM

ITESM Ciudad de México

In recognition of your achievement as the **Winner** of the **Data and Telemetry Award sponsored by Schmid Elektronik** at Shell Eco-marathon.

Norman Koch
Global General Manager
Shell Eco-marathon
April 06, 2025

40 years
Shell Eco-marathon
1985–2025

Shell
Eco-marathon

Americas 2025

Certificate of Results

Presented to:

Paulina Lilian Ruíz Servín

EcoVolt CCM

ITESM Ciudad de México

In recognition of your exceptional work in vehicle design, testing and sucessful completion of Technical Inspection.

Norman Koch
Global General Manager
Shell Eco-marathon
April 06, 2025

40 years
Shell Eco-marathon
1985–2025

Shell
Eco-marathon

Brazil 2025

Certificate of Results

Presented to:

Paulina Lilian Ruíz Servín

EcoVolt CCM

IESM Ciudad de México

In recognition of your exceptional work in vehicle design, testing and sucessful completion of Technical Inspection.

Norman Koch
Global General Manager
Shell Eco-marathon
August 28, 2025

40 years
Shell Eco-marathon
1985–2025

Shell
Eco-marathon

United States 2026

Off-Track Award Certificate

Presented to:

EcoVolt CCM

Instituto Tecnológico y de Estudios Superiores de Monterrey Campus Ciudad de México

In recognition of your achievement as the **Runner-Up** of the **Data and Telemetry Award sponsored by Schmid Elektronik** at Shell Eco-marathon.

Norman Koch
Global General Manager
Shell Eco-marathon
April 10, 2026

40 years
Shell Eco-marathon
1985–2025

Shell
Eco-marathon

United States 2026

Off-Track Award Certificate

Presented to:

EcoVolt CCM

Instituto Tecnológico y de Estudios Superiores de Monterrey Campus Ciudad de México

In recognition of your achievement as the Winner of the Vehicle Design Award (Prototype) sponsored by Qatar Museums at Shell Eco-marathon.

Norman Koch
Global General Manager
Shell Eco-marathon
April 10, 2026

40 years
Shell Eco-marathon
1985–2025

Shell
Eco-marathon

United States 2026

Certificate of Results

Presented to:

Paulina Lilian Ruíz Servín

EcoVolt CCM

Instituto Tecnológico y de Estudios Superiores de Monterrey Campus Ciudad de México

In recognition of your exceptional work in vehicle design, testing and sucessful completion of Technical Inspection.

Norman Koch
Global General Manager
Shell Eco-marathon
April 10, 2026

40 years
Shell Eco-marathon
1985–2025

Shell
Eco-marathon

Brazil 2026

Certificate of Participation

Presented to:

Paulina Lilian Ruíz Servín
EcoVolt CCM
Instituto Tecnológico y de Estudios Superiores de Monterrey Campus Ciudad de México

In recognition of your participation in Shell Eco-marathon. Your efforts in vehicle design, development, and teamwork contribute to the spirit of innovation and learning that define the competition.

Norman Koch
Global General Manager
Shell Eco-marathon
August 27, 2026